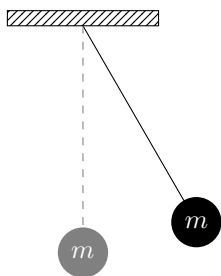


Chapter 16A Homework (Simple Harmonic Motion)



Homework (collected on Mini-Test Day: Fri, Feb 13)

Reading

Please read the following on your own in the OpenStax textbook by the dates given. It will give good context for class discussion. Check off when you have completed them.

- ☐ 16.1 Hooke's Law Fri, Feb 27
- ☐ 16.2 Period & Frequency Fri, Feb 27
- ☐ 16.4 The Simple Pendulum Fri, Mar 6
- ☐ 16.7 Force Oscillations & Resonance Fri, Mar 6

Problems and Conceptual Question

Get stamps from your instructor as you complete each of the following problems. The conceptual questions (CQ) require at least one sentence of explanation.

STAMPS WILL NOT BE GIVEN IF WORK IS NOT SHOWN ON A SEPARATE SHEET OF PAPER

Springs (8 POINTS) Problems #1-3 Conceptual #5-10	Equations of Motion (3 POINTS) Problems #10-12
Energy in SHM (4 POINTS) Problems #14-16	Simple Pendulum (5 POINTS) Problems #17-19 Conceptual #20-21

Bonus Problem

#4

Equations

$$F = -kx$$

$$KE = \frac{1}{2}mv^2$$

$$PE_e = \frac{1}{2}kx^2$$

$$f_s = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

$$f_p = \frac{1}{2\pi}\sqrt{\frac{g}{L}}$$

$$\omega = 2\pi f$$

$$x(t) = A \cos(\omega t + \phi)$$

$$v(t) = -v_{\max} \sin(\omega t + \phi)$$

$$v_{\max} = A\omega$$

Selected Answers

1. 7.54 cm
2. 400 Hz
3. 94.7 kg
5. 0.84 m
6. at equilibrium
7. $4m$
8. never
9. A, C, and D
10. B
11. 2.25 m & 3.5 m; 0.25 Hz & 0.5 Hz; 4 s & 2 s
12. 0.23 sec
13. $x = 0.28 \cos(36.0t + 3\pi/2)$; $t = 0.04$ s & 0.13 s
14. 0.650 m; 1.34 Hz; Total = 17.2 J; $PE = 5.26$ J; $KE = 11.92$ J
15. 5.5 cm; 0.59 m/s
16. (a) 0.335 m/s; (b) 5.61×10^{-4} J
17. 3.0 sec
18. 0.994 m
19. 1.8 sec; 0.56 Hz
20. lengthen
21. period gets cut by $1/\sqrt{2}$

Questions and figures are from OpenStax *College Physics*, 2nd ed, (Chapter 16) or Giancoli *Physics: Principles with Applications*, 7th ed, (Chapter 11) unless otherwise noted. unless otherwise noted.

Springs and Hooke's Law

Problems

- (P4a) The springs of a pickup truck act like a single spring with a force constant of $1.30 \times 10^5 \text{ N/m}$. By how much will the truck be depressed by its maximum load of 1000 kg?
- (P9) Find the frequency of a tuning fork that takes $2.50 \times 10^{-3} \text{ s}$ to complete one oscillation.
- (P18) A diver on a diving board is undergoing simple harmonic motion. Her mass is 55.0 kg and the period of her motion is 0.800 s. The next diver is a male whose period of simple harmonic oscillation is 1.05 s. What is his mass if the mass of the board is negligible?
- Bonus** (Giancoli P23) A bungee jumper with mass 65.0 kg jumps from a high bridge. After arriving at his lowest point, he oscillates up and down, reaching a low point seven more times in 43.0 s. He finally comes to rest 25.0 m below the level of the bridge. Estimate the spring stiffness constant and the unstretched length of the bungee cord assuming SHM.

Conceptual Questions

- (Giancoli P1) If a particle undergoes SHM with amplitude 0.21 m, what is the total distance it travels in one period?
- (Giancoli CQ1) Is the acceleration of a simple harmonic oscillator ever zero? If so, where?
- (Giancoli MC3) An object of mass m oscillates on the end of a spring. To double the period, you should replace the object with a mass of what?
- (Giancoli MQ1) A mass on a spring in SHM (Figure 1) has amplitude A and period T . At what point in the motion (if any) is the velocity zero and the acceleration zero simultaneously?

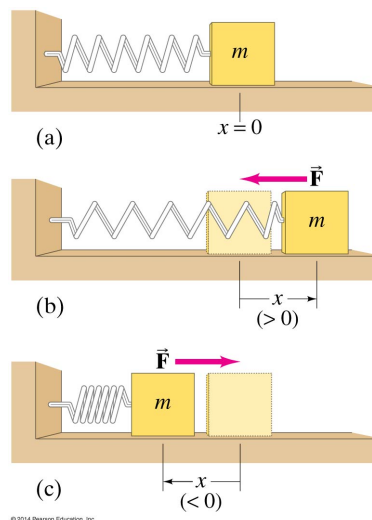


Figure 1: Giancoli Figure 11-01.

- (Giancoli MC2) An object oscillates back and forth on the end of a spring. Which of the following statements are true at some time during the course of the motion?
 - The object can have zero velocity and, simultaneously, nonzero acceleration.
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 - The object can have zero acceleration and, simultaneously, nonzero velocity.
 - The object can have nonzero velocity and nonzero acceleration simultaneously.
- (Giancoli MC4) An object of mass m rests on a frictionless surface and is attached to a horizontal ideal spring with spring constant k . The system oscillates with amplitude A . The oscillation frequency of this system can be increased by which of the following?
 - decreasing k .
 - decreasing m .
 - increasing A .
 - More than one of the above.
 - None of the above will work.

Equations of Motion

Problems

11. (Giancoli P9) Figure 2 shows two examples of SHM, labeled A and B. For each, what is (a) the amplitude, (b) the frequency, and (c) the period?

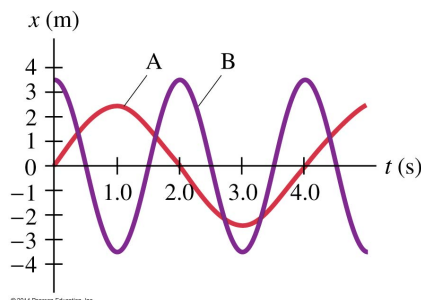


Figure 2: Giancoli Figure 11-51.

12. (Giancoli P13) A 1.65-kg mass stretches a vertical spring 0.215 m. If the spring is stretched an additional 0.130 m and released, how long does it take to reach the (new) equilibrium position again?
13. (Giancoli P8) A vertical spring with spring stiffness constant 305 N/m oscillates with an amplitude of 28.0 cm when 0.235 kg hangs from it. The mass passes through the equilibrium point ($y = 0$) with positive velocity at $t = 0$, which means that $\phi = 3\pi/2$. (a) What equation describes this motion as a function of time? (b) At what times will the spring be longest and shortest?

Energy in SHM

Problems

14. (Giancoli P14) A 1.15-kg mass oscillates according to the equation

$$x = 0.650 \cos(8.4t)$$

where x is in meters and t in seconds. Determine (a) the amplitude, (b) the frequency, (c)

the total energy, and (d) the kinetic energy and potential energy when $x = 0.36$ m.

15. (Giancoli P20) An object with mass 2.7 kg is executing simple harmonic motion, attached to a spring with spring constant $k = 310$ N/m. When the object is 0.020 m from its equilibrium position, it is moving with a speed of 0.55 m/s. (a) Calculate the amplitude of the motion. (b) Calculate the maximum speed attained by the object.
16. (P38) A novelty clock has a 0.0100-kg mass object bouncing on a spring that has a force constant of 1.25 N/m. What is the maximum velocity of the object if the object bounces 3.00 cm above and below its equilibrium position? (b) How many joules of kinetic energy does the object have at its maximum velocity?

Simple Pendulum

Problems

17. (Giancoli P27) A pendulum has a period of 1.85 s on Earth. What is its period on Mars, where the acceleration of gravity is about 3.63 m/s²?
18. (Giancoli P28) How long must a simple pendulum be if it is to make exactly one swing per second? (That is, one complete oscillation takes exactly 2.0 s.)
19. (Giancoli P29) A pendulum makes 28 oscillations in exactly 50 s. What is its (a) period and (b) frequency?

Conceptual Questions

20. (CQ8) Pendulum clocks are made to run at the correct rate by adjusting the pendulum's length. Suppose you move from one city to another where the acceleration due to gravity is slightly greater, taking your pendulum clock with you, will you have to lengthen or shorten the pendulum to keep the correct time, other factors remaining constant? Explain your answer.
21. (P30) What is the effect on the period of a pendulum if you double its length?